Python for Data Science & AI

YOUTUBE PRESENTATION

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FIRST STEPS

```python
youtube_categories = {
    1: "Film & Animation",
    2: "Autos & Vehicles",
    10: "Music",
    15: "Pets & Animals",
    20: "Gaming",
    22: "People & Blogs",
    23: "Comedy",
    24: "Entertainment",
    25: "News & Politics",
    26: "Howto & Style",
    28: "Science & Technology",
    29: "Nonprofits & Activism"
}
```

```python
import pandas as pd
from googleapiclient.discovery import build

api_key = "AIzaSyCs6dW1VHrhSa7zuKC3TREhzdA7Y_P3_5I"
youtube = build('youtube', 'v3', developerKey=api_key)

video_data = []

for category_id, category_name in youtube_categories.items():
  request = youtube.videos().list(
    part="contentDetails,snippet,statistics",
    chart="mostPopular",
    regionCode="US",
    videoCategoryId=category_id,
    maxResults=50
  )
  response = request.execute()

  for video in response['items']:
    title = video['snippet']['title']
    channel = video['snippet']['channelTitle']
    channel_id = video['snippet']['channelId']
    duration = video['contentDetails']['duration']
    views = int(video['statistics']['viewCount'])
    likes = int(video['statistics'].get('likeCount', 0))

    video_data.append({
        'Title': title,
        'Channel': channel,
        'Channel_ID': channel_id,
        'Duration': duration,
        'Views': views,
        'Likes': likes,
        'Category': category_name
    })

df_videos = pd.DataFrame(video_data)
df_videos
```

	Title	Channel	Channel_ID	Duration	Views	Likes	Category
0	If you win the game,I'll let you go.#movie #sh...	AllAccess TV	UCkb80lSeHu2n4lA6dwdRQ5g	PT58S	9569924	667905	Film & Animation
1	"Get In The Car"|Bad Boys: Ride or Die (2024)|...	Clinjo AE	UCgcX8KN9tjesfSg1KYW8JMA	PT56S	11473664	0	Film & Animation
2	His parents were surprised by his change.#shorts	PureVision	UCFCiuH9RyKztPPi6hoTm0NQ	PT1M16S	7049438	509032	Film & Animation
3	This show is so good	MovieLuxeShorts	UCxcwb1pqg2BtlR1AWSEX-MA	PT56S	6425641	668536	Film & Animation
4	This is a wonderful scene #shorts	Reel Spark	UCktj7by7UzpVLP7INSgo3ZQ	PT58S	6109577	306462	Film & Animation
...	...	...	...	...	...	...	...
526	Introducing the "VitaWear SmartBand," a next-g...	BB DNSIER	UCtfH9NuEOg6VnyoloxORTqQ	PT47S	2308127	191435	Science & Technology
527	iPhone vs android in the Future 😂	Speed McQueen	UCNFT_eq_QCApMEwCACHto9Q	PT19S	2589306	191388	Science & Technology
528	Don't make this mistake	Matty McTech	UC51UPs8iqgC6ugVROBcUQEQ	PT46S	246865	14605	Science & Technology
529	We need your help!	Ben Esherick	UCFXCGN9-roeqdFAatAwG8ew	PT54S	544271	70568	Nonprofits & Activism
530	Joe Rogan On This American Organization. JRE P...	Mata Kanan	UCsuPzSnvL1ak634rMVEtcPw	PT19S	87812	0	Nonprofits & Activism

531 rows × 7 columns

Get top 50 videos of each categories with columns shown above containing information about the videos that we require

EXPLANATORY
DATA ANALYSIS

```
df_videos.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 532 entries, 0 to 531
Data columns (total 7 columns):
 #   Column      Non-Null Count  Dtype
---  ------      --------------  -----
 0   Title       532 non-null    object
 1   Channel     532 non-null    object
 2   Channel_ID  532 non-null    object
 3   Duration    532 non-null    object
 4   Views       532 non-null    int64
 5   Likes       532 non-null    int64
 6   Category    532 non-null    object
dtypes: int64(2), object(5)
memory usage: 29.2+ KB
```

```
df_videos.describe()
```

	Views	Likes
count	5.320000e+02	5.320000e+02
mean	5.441110e+06	2.322374e+05
std	7.865160e+06	3.257376e+05
min	1.651900e+04	0.000000e+00
25%	6.535015e+05	1.766675e+04
50%	2.570142e+06	1.010100e+05
75%	7.143824e+06	3.245148e+05
max	7.479083e+07	2.622824e+06

```
df_videos.shape

(532, 7)
```

HYPOTHESIS 1

- **Null hypothesis**: There is no significant relationship between a channel's subscriber count and its 'Activity Index' (calculated as total videos divided by channel age).
- **Alternative hypothesis**: There is a significant positive relationship between a channel's subscriber count and its 'Activity Index' (calculated as total videos divided by channel age).

```python
unique_channels = df_videos['Channel_ID'].unique().tolist()

channel_data = []

unique_channels = [channel_id for channel_id in unique_channels if channel_id]

for i in range(0, len(unique_channels), 50):
    request = youtube.channels().list(
        part="snippet,statistics",
        id=",".join(unique_channels[i:i+50])
    )
    response = request.execute()

    for item in response['items']:
        channel_data.append({
            'Channel_ID': item['id'],
            'Channel_Title': item['snippet']['title'],
            'PublishedAt': item['snippet']['publishedAt'],
            'Subscribers': int(item['statistics'].get('subscriberCount', 0)),
            'VideoCount': int(item['statistics'].get('videoCount', 0))
        })

df_channels = pd.DataFrame(channel_data)
df_channels
```

	Channel_ID	Channel_Title	PublishedAt	Subscribers	VideoCount
0	UCQb2BjZ9xyXfHfgUyIF463A	Primo Brickhouse	2022-10-06T11:57:05.858158Z	40100	1510
1	UCBguopg2BEArRruRyEPqOgw	Sanzecut	2024-10-16T04:21:41.384436Z	73400	87
2	UC6sFP8srPDw7-3F-d10XYZQ	Comedy Currency	2025-08-26T03:06:32.289373Z	17500	26
3	UCkJWHMoAIMM5eZJtfMfyCvg	Global MIR4 Shorts	2018-10-21T03:54:15Z	30600	130
4	UCAnPqNomicDKFV4GVOkBBOw	PunchLine Comedy	2019-01-22T00:33:44Z	32800	354
...	...	...	...	...	...
442	UCuwJoiGWRxPYStBp0I0WuZw	Luke Davidson	2012-01-09T19:05:33Z	17600000	1757
443	UCNFT_eq_QCApMEwCACHto9Q	Speed McQueen	2018-09-16T14:21:05Z	11000000	5017
444	UCPvRdfUooqCf0zPOYvbBzVA	mryeester	2020-02-27T19:43:10.612886Z	1630000	995
445	UCClfFsWcT3N2I7VTXXyt84A	Salem Techsperts	2014-10-15T21:30:10Z	2290000	500
446	UCy0tKL1T7wFoYcxCe0xjN6Q	Technology Connections	2014-11-09T21:54:10Z	2910000	221

447 rows × 5 columns

Get top 50 unique channels of each categories with columns shown above containing its ID and name, age of channel, subscriber count and video count

```
from datetime import datetime

df_channels['PublishedAt'] = pd.to_datetime(df_channels['PublishedAt'], format='mixed')
df_channels['FormattedDate'] = df_channels['PublishedAt'].dt.strftime('%Y-%m-%d')

today = pd.Timestamp(datetime.now().date())
df_channels['ChannelAge'] = (today - pd.to_datetime(df_channels['FormattedDate'])).dt.days // 30

df_channels

df_cleaned = df_channels[df_channels['ChannelAge'] != 0]
df_cleaned
```

	Channel_ID	Channel_Title	PublishedAt	Subscribers	VideoCount	FormattedDate	ChannelAge
0	UCQb2BjZ9xyXfHfgUyIF463A	Primo Brickhouse	2022-10-06 11:57:05.858158+00:00	40100	1510	2022-10-06	36
1	UCBguopg2BEArRruRyEPqOgw	Sanzecut	2024-10-16 04:21:41.384436+00:00	73400	87	2024-10-16	12
2	UC6sFP8srPDw7-3F-d10XYZQ	Comedy Currency	2025-08-26 03:06:32.289373+00:00	17500	26	2025-08-26	1
3	UCkJWHMoAIMM5eZJtfMfyCvg	Global MIR4 Shorts	2018-10-21 03:54:15+00:00	30600	130	2018-10-21	84
4	UCAnPqNomicDKFV4GVOkBBOw	PunchLine Comedy	2019-01-22 00:33:44+00:00	32800	354	2019-01-22	81
...	...	...	...	...	...	...	...
442	UCuwJoiGWRxPYStBp0l0WuZw	Luke Davidson	2012-01-09 19:05:33+00:00	17600000	1757	2012-01-09	167
443	UCNFT_eq_QCApMEwCACHto9Q	Speed McQueen	2018-09-16 14:21:05+00:00	11000000	5017	2018-09-16	86
444	UCPvRdfUooqCf0zPOYvbBzVA	mryeester	2020-02-27 19:43:10.612886+00:00	1630000	995	2020-02-27	68
445	UCClfFsWcT3N2I7VTXXyt84A	Salem Techsperts	2014-10-15 21:30:10+00:00	2290000	500	2014-10-15	133
446	UCy0tKL1T7wFoYcxCe0xjN6Q	Technology Connections	2014-11-09 21:54:10+00:00	2910000	221	2014-11-09	133

437 rows × 7 columns

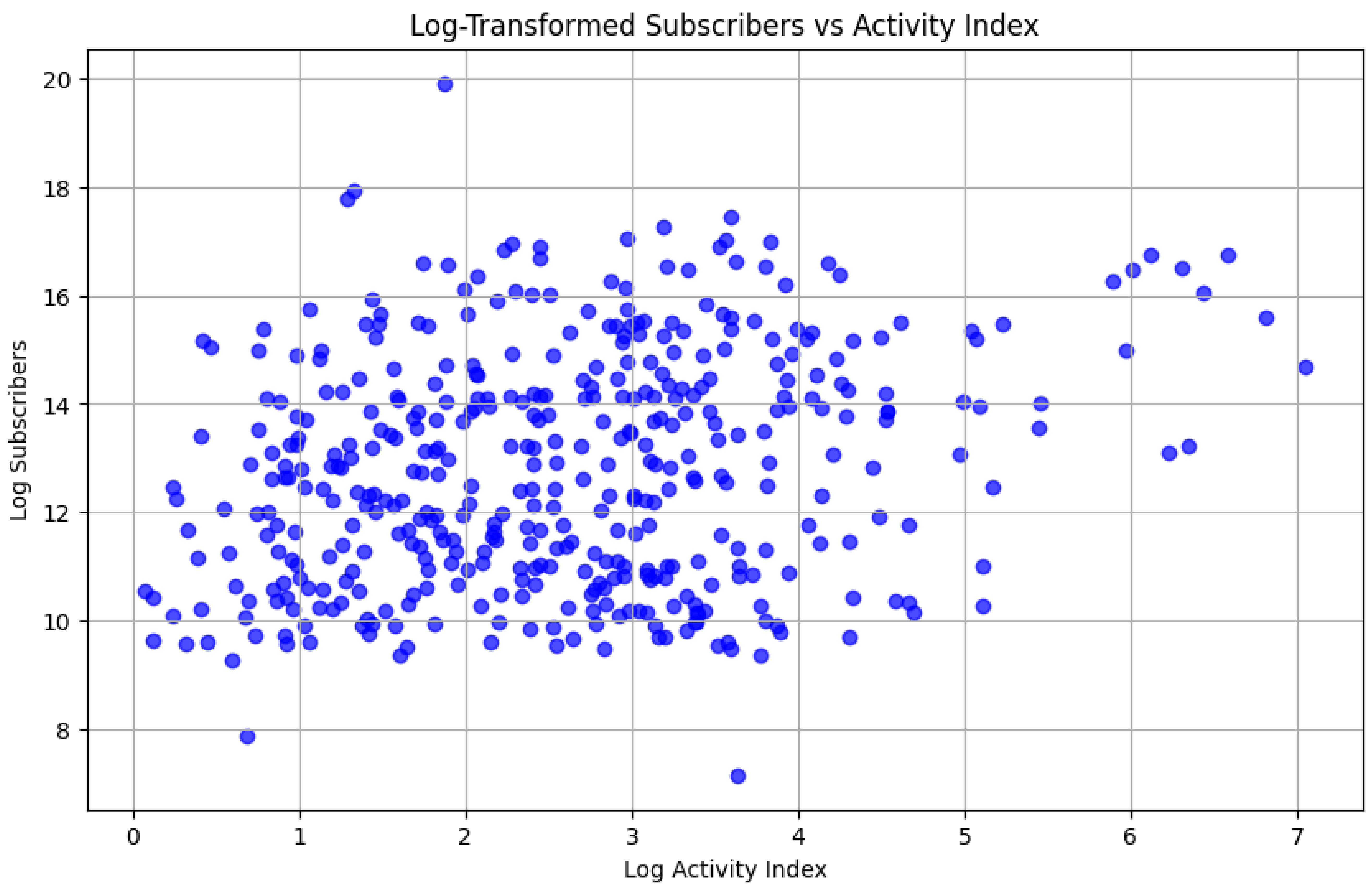
```python
import pandas as pd
import matplotlib.pyplot as plt
import numpy as np

df_cleaned['ActivityIndex'] = df_cleaned['VideoCount'] / df_cleaned['ChannelAge']

df_cleaned['log_ActivityIndex'] = np.log1p(df_cleaned['ActivityIndex'])
df_cleaned['log_Subscribers'] = np.log1p(df_cleaned['Subscribers'])

plt.figure(figsize=(10,6))
plt.scatter(df_cleaned['log_ActivityIndex'], df_cleaned['log_Subscribers'], color='blue', alpha=0.7)
plt.xlabel('Log Activity Index')
plt.ylabel('Log Subscribers')
plt.title('Log-Transformed Subscribers vs Activity Index')
plt.grid(True)
plt.show()
```

Show the normalized scatter plot data with Y axis being Subscribers and X axis being Activity Index


Log-Transformed Subscribers vs Activity Index
Log Subscribers
Log Activity Index
20
18
16
14
12
10
8
0
1
2
3
4
5
6
7

RESULTS

```
from scipy.stats import pearsonr

corr, p_value = pearsonr(df_cleaned['ActivityIndex'], df_cleaned['Subscribers'])

print(f"Pearson correlation: {corr}")
print(f"P-value: {p_value}")

if p_value < 0.05:
    print("Reject H₀: There is enough evidence to suggest that ActivityIndex significantly affects Subscribers.")
else:
    print("Fail to reject H₀: There is not enough evidence to suggest that ActivityIndex affects Subscribers.")
```

Pearson correlation: 0.042843739811802714

P-value: 0.371603416767697606

Fail to reject H_0: There is not enough evidence to suggest that Activity Index affects Subscribers.

HYPOTHESIS 2

- **Null hypothesis**: The most frequently used keywords (top 30) in a video title has no significant relationship with video views.
- **Alternative hypothesis**: The utilization of the most frequently used keywords (top 30) in a video title gives videos more views.

```python
df_2 = df_videos.sort_values(by='Views', ascending=False)
all_titles = " ".join(df_2['Title'].str.lower())
df_2
```

	Title	Channel	Channel_ID	Duration	Views	Likes	Category
369	100 Kids Vs World's Strongest Man!	MrBeast	UCX6OQ3DkcsbYNE6H8uQQuVA	PT27M21S	49052287	1518429	Entertainment
100	Taylor Swift - The Fate of Ophelia (Official M...	Taylor Swift	UCqECaJ8Gagnn7YCbPEzWH6g	PT3M59S	47565944	1335464	Music
171	THIS WAS THE BEST ONE YET! 😆	Double Date	UCoJ5osZ535ar2kzHwQMnLsA	PT33S	42686030	384015	Pets & Animals
68	how to resolve cars like this #driving #truckd...	Driver Lover	UCxb5y7usqPLCxo-sBYNBiiw	PT1M2S	38843749	2521945	Autos & Vehicles
16	«How you doin?» 😂 | Norbit- Rasputia going down ...	Ztick Fish edits	UCWCM0kLI1gy93GKk49Z0bpw	PT56S	30008418	908248	Film & Animation
...	...	...	...	...	...	...	...
263	This just saved me $30 at least by not swingin...	Beyondbeautybyshannon	UC-R3etkvm1fzXobn7-ntH6w	PT1M10S	31085	3229	People & Blogs
247	💥 Would Civil War over THERE Break One Loose H...	Appalachia's Homestead with Patara	UCXyRtDXzptYN56-UxNhYAbQ	PT20M6S	24214	3568	People & Blogs
264	week in my life 🎀 🪞 👯 apartment decor, cooking a...	Danielle Marie Carolan	UCiGWpX9oCmMu3hRUV_ZBvpg	PT29M55S	22619	711	People & Blogs
246	Matt usually completely unpacks for me because...	Jessica	UCxPtWxbojpMGbpQKXPthzuw	PT2M59S	21741	4517	People & Blogs
256	PLANNING OUR HONEYMOON!	Brock and Boston	UCEM_-cD5W5fltaAt_LITgwg	PT26M46S	20184	1087	People & Blogs

531 rows × 7 columns

```
import re
from collections import Counter

words = re.findall(r'\b[a-z]{3,}\b', all_titles)
common_words = Counter(words).most_common(50)
print(common_words)
```

- the: 123
- shorts: 88
- this: 47
- you: 31
- and: 30
- for: 27
- video: 24
- how: 23
- with: 21
- car: 19
- official: 18
- she: 18
- cat: 18
- funny: 18
- his: 17
- new: 16
- music: 14
- was: 14
- trump: 14
- viral: 13
- baby: 12
- fyp: 12
- ranking: 12
- are: 12
- edit: 11
- not: 11
- they: 11
- just: 11
- one: 10
- her: 10
- after: 10
- into: 10
- day: 10
- news: 10
- steal: 10
- like: 9
- got: 9
- family: 9
- when: 9
- can: 9
- but: 9
- moments: 9
- home: 9
- our: 9
- cars: 8
- real: 8
- cats: 8
- bear: 8
- get: 8
- ever: 8

```python
from nltk.corpus import stopwords
import nltk
nltk.download('stopwords')

stop_words = set(stopwords.words('english'))

clickbait_words = [(w) for w, c in common_words if w not in stop_words]
print(len(clickbait_words))
```

```python
def contains_clickbait(title):
    title = title.lower()
    return any(word in title for word in clickbait_words)

df_2['Has_Clickbait'] = df_2['Title'].apply(contains_clickbait)
df_2.head()
```

	Title	Channel	Channel_ID	Duration	Views	Likes	Category	Has_Clickbait
369	100 Kids Vs World's Strongest Man!	MrBeast	UCX6OQ3DkcsbYNE6H8uQQuVA	PT27M21S	49052287	1518429	Entertainment	False
100	Taylor Swift - The Fate of Ophelia (Official M...	Taylor Swift	UCqECaJ8Gagnn7YCbPEzWH6g	PT3M59S	47565944	1335464	Music	True
171	THIS WAS THE BEST ONE YET! 😆	Double Date	UCoJ5osZ535ar2kzHwQMnLsA	PT33S	42686030	384015	Pets & Animals	True
68	how to resolve cars like this #driving #truckd...	Driver Lover	UCxb5y7usqPLCxo-sBYNBiiw	PT1M2S	38843749	2521945	Autos & Vehicles	True
16	«How you doin?» 😂 | Norbit- Rasputia going down ...	Ztick Fish edits	UCWCM0kLI1gy93GKk49Z0bpw	PT56S	30008418	908248	Film & Animation	True

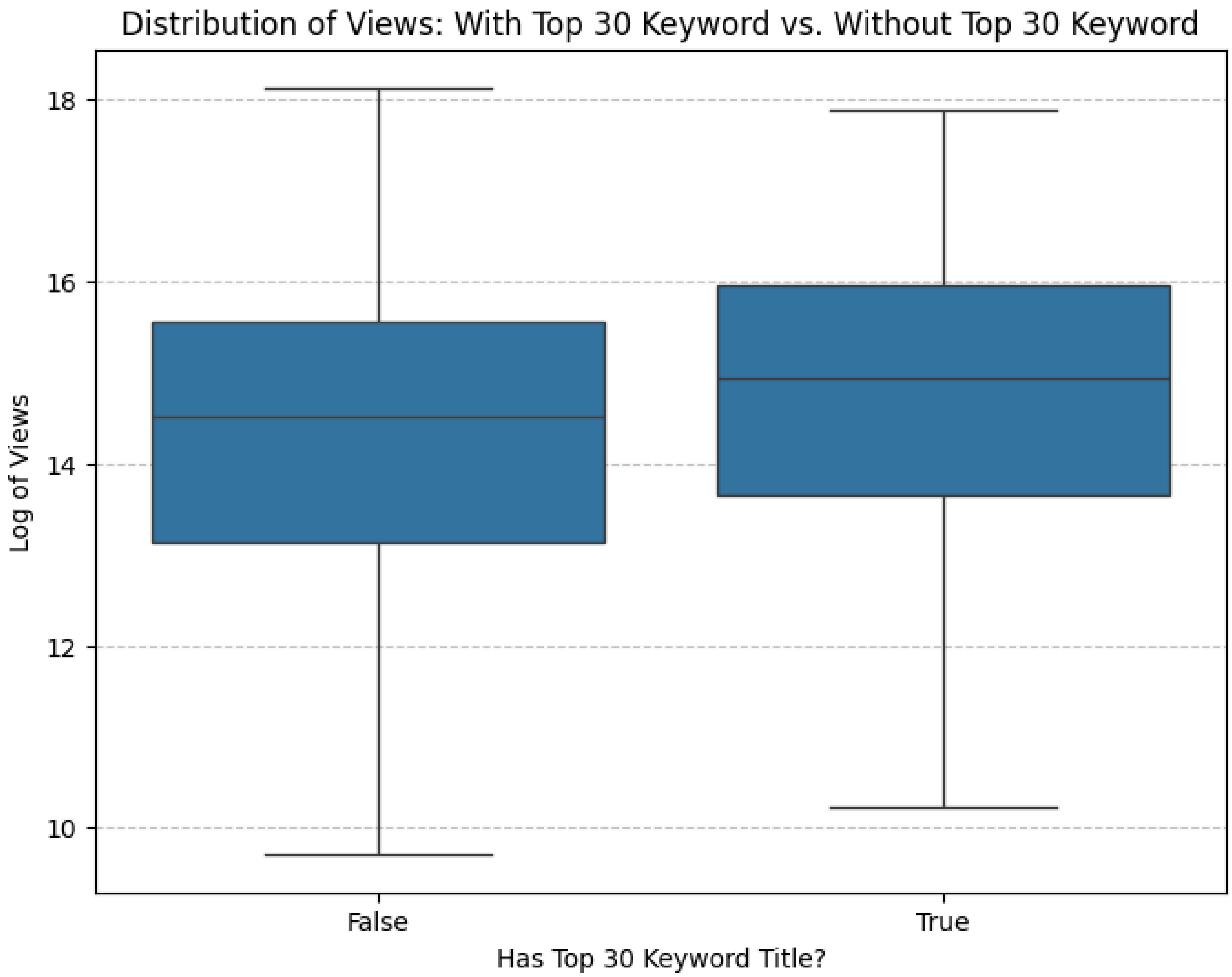
```python
import seaborn as sns
import matplotlib.pyplot as plt
import numpy as np

df_2['log_Views'] = np.log1p(df_2['Views'])

plt.figure(figsize=(8, 6))
sns.boxplot(x='Has_Clickbait', y='log_Views', data=df_2)

plt.title('Distribution of Views: With Top 30 Keyword vs. Without Top 30 Keyword')
plt.xlabel('Has Top 30 Keyword Title?')
plt.ylabel('Log of Views')
plt.grid(axis='y', linestyle='--', alpha=0.7)
plt.show()
```

Show the normalized boxplot data with Y axis being Views and X axis being information on whether the title has top 30 keywords or not



Distribution of Views: With Top 30 Keyword vs. Without Top 30 Keyword
18
16
14
12
10
Log of Views
False
True
Has Top 30 Keyword Title?

RESULTS

```python
from scipy.stats import pearsonr

corr, p_value = pearsonr(df_2['Has_Clickbait'], df_2['Views'])

print(f"Pearson correlation: {corr}")
print(f"P-value: {p_value}")

if p_value < 0.05:
    print("Reject H₀: There is enough evidence to suggest that most frequently used keywords in title positively affects video views.")
else:
    print("Fail to reject H₀: There is not enough evidence to suggest that most frequently used keywords in title positively affects video views.")
```

Pearson correlation: 0.10691301351370505
P-value: 0.013704922309818474
Reject H_0: There is enough evidence to suggest that the utilization of the most frequently used keywords (top 30) in a video title gives videos more views.

Python for Data Science & AI

THANK YOU

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REFERENCE

YouTube Data API | Google for Developers. (n.d.). Google for Developers. Retrieved October 1, 2025, from https://developers.google.com/youtube/v3